

Functional Requirements Document (FRD)

Introduction

This Functional Requirements Document (FRD) is derived from the Business Requirements Document (BRD) provided. The purpose of this document is to outline the functional requirements of the "Application" that is designed to handle text normalization and metadata processing from various sources.

Functional Requirements

1. **Text Normalization Module**:

- The Application shall provide a text normalization functionality that can convert text to a standard format.
- The normalization functionality shall include but not be limited to:
 - Converting text to lowercase.
 - Removing or handling special characters.
 - Handling non-English characters.

2. **Metadata Extraction and Processing Module**:

- The Application shall be capable of extracting metadata from various file types and data sources, including:
 - Excel files.
 - XML documents.
 - ETL (Extract, Transform, Load) mappings.
 - Datasets.
 - System configuration files.
 - Log files.
- The Application shall process the extracted metadata.

3. **Flexibility and Scalability**:

- The Application shall be designed to be flexible and scalable to handle different types and sizes of metadata.

4. **Data Integrity and Consistency**:

- The Application shall ensure that the processed data maintains integrity and consistency, especially when dealing with large datasets or complex metadata.

User Interface Requirements

1. **Upload/Input Metadata**:

- The Application shall provide an intuitive interface for users to upload or input metadata from different sources.

2. **Configure Processing Options**:

- The Application shall allow users to configure processing options for text normalization and metadata extraction.

3. **View Results**:

- The Application shall provide a functionality to view the results of the text normalization and metadata processing.

System Requirements

1. **Performance**:

- The Application shall be able to process metadata efficiently and handle large volumes of data without significant performance degradation.

2. **Security**:

- The Application shall ensure the security of the processed data and adhere to relevant data protection standards and regulations.

3. **Compatibility**:

- The Application shall be compatible with various data sources and file formats to ensure broad applicability.

Data Flow Diagram

To visualize the data flow within the Application, we will use the `generate_dfd_diagram` tool with the description: "Data Flow Diagram for Text Normalization and Metadata Processing Application, showing the flow from input metadata sources through text normalization and metadata extraction to processed data output."`

```
```python
```

```
generate_dfd_diagram(description="Data Flow Diagram for Text Normalization and Metadata Processing Application, showing the flow from input metadata sources through text normalization and metadata extraction to processed data output.")
```

```
```
```

Acceptance Criteria

The Application shall meet the following acceptance criteria:

1. Successfully normalize text data from the input metadata.
2. Extract and process metadata from a variety of file types and data sources.

3. Maintain data integrity and consistency during processing.
4. Perform well under large data volumes.
5. Ensure security and adherence to relevant data protection regulations.

Conclusion

This FRD outlines the functional requirements for the Application to handle text normalization and metadata processing. It provides a detailed overview of the necessary functionalities, user interface, and system requirements to meet the business needs outlined in the BRD.